简体中文 | English

周彤昕

13396210825 | zhoutx8@mail2.sysu.edu.cn

基本信息

性别: 男　**出生年月:** 2003年02月　**籍贯:** 山东省菏泽市

本科院校: 中山大学集成电路学院（学院首届本科生）　**政治面貌:** 中共党员

专业: 微电子科学与工程（2021级）　**GPA:** 86.17　**年级排名:** 9/87（15%）

- **外语水平**：**CET-4(550), CET-6(485)**，CET-6**口语良好**，具备优异的英语**听说读写**和**沟通交际**能力；
- **主修课程**：程序设计、线性代数、高等数学、理论力学、复变函数、大学物理、模拟电路、数字电路、数学物理方法、信号与系统、模拟集成电路设计、数字集成电路设计、量子力学与固体物理、电动力学、半导体物理、微机原理与嵌入式系统、人工智能与硬件加速、先进微纳电子器件、数字信号处理、电子设计自动化等；
- **兴趣方向**：人工智能与硬件加速方案、集成电路芯片架构（**FPGA**）、高性能深度学习模型训推算法（**Python**）、模拟集成电路与系统设计等。

荣誉获奖

- 2024年11月，获**中山大学优秀学生三等奖学金**；
- 2024年09月，获**芯火科创协会优秀学生社团骨干**；
- 2024年08月，获**第八届全国大学生集成电路创新创业大赛华南赛区二等奖**；
- 2023年12月，获**国家奖学金**；
- 2023年11月，获**中山大学优秀学生一等奖学金**；
- 2023年11月，获**中山大学职业规划大赛院赛一等奖**；
- 2023年11月，获**中山大学暑期文化科技卫生“三下乡”社会实践优秀个人**；
- 2023年08月，获**全国大学生电子设计竞赛广东赛区二等奖**；
- 2022年11月，获**中山大学专项奖学金**；
- 2022年10月，获**全国大学生数学建模竞赛广东省分赛二等奖**。

科研经历

基于参数融合的中文大语言模型构建与优化研究（本科毕业设计）

- **项目概况**：**大语言模型（LLM）**的**参数融合**已经发展成为一种高效的模型构建方法，寻求**中文大语言模型**的融合成果；
- **主要工作**：在导师的指导下，负责实施模型融合实验，**独立**进行性能评估与优化策略的迭代；
- **研究成果**：目前已经完成多组模型的融合测试，正在逐步**优化模型性能**；

LAV: 基于LoongArch架构的R-MSFM模型专用加速器设计（第八届全国大学生集成电路创新创业大赛）

- **项目概况**：基于**龙芯架构**提出了一个面向**专用神经网络**高度定制化的视觉网络**模型推理加速器**硬件架构（LA-Vision）；
- **主要工作**：承担**代码实现**、**电路设计**与**仿真验证**工作，主导**板卡调试**与**现场展示**；
- **研究成果**：通过华南赛区初筛，在现场展示后获得**第八届全国大学生集成电路创新创业大赛华南赛区二等奖**；

面向机器人视觉系统的深度估计硬件处理器设计（大学生创新训练项目）

- **项目概况**：面向**机器人视觉应用**，基于**FPGA**设计适用于搭载机器学习**单目深度估计算法的硬件处理器**；
- **主要工作**：模型的**量化策略**测试与选定、**硬件加速**方案设计等；
- **研究成果**：获**校级立项**，已成功构建**多层深度估计硬件处理器**，目前致力于**实时数据处理**，计划实现**边端实时深度估计**；

运动目标控制与追踪系统（全国大学生电子设计竞赛）

- **课题概况**：2023年中国大学生电子设计竞赛E题，设计一个系统，通过按键控制实现**复位**、**绕线**、**循迹**、**追踪**等功能；
- **主要工作**：负责**系统程序开发**、**电路设计与测试**；
- **最终成果**：通过不同的按键操作可以实现题目所要求的功能。装置在赛区会场进行测试，与报告共同获得**广东赛区二等奖**。

基于K-均值聚类的古代玻璃制品化学成分分析模型（全国大学生数学建模竞赛）

- **课题概况**：2022 全国大学生数学建模竞赛C题，要求通过建模分析与古代玻璃化学成分相关的数据；
- **主要工作**：担任项目组长，负责**建模原理研究**、**算法实现**；
- **最终成果**：解决赛题中所提出的化学成分分析问题，与报告共同获得**广东省二等奖**；

面向MEMS电容式加速度计的低功耗高分辨率Delta-Sigma型调制器（大学生创新训练项目）

- **课题概况**：2022年完成题目，研究满足MEMS电容式**精度**要求且**功耗**为微瓦级的Delta-Sigma技术；
- **主要工作**：作为项目负责人，主导**参数建模**、**电路设计与优化**，以及撰写结项报告并进行**答辩**；
- **最终成果**：题目结项为合格，最终调制器**有效位数16比特**，**功耗为1.67微瓦**，**量化信噪比101分贝**，具备一定的应用价值。

兴趣技能

- **个人技能**：C/Python/Markdown、cadenceEDA套件、AutoCAD、Altium Designer、Vivado、Proteus等；
- **兴趣爱好**：视频制作、新媒体运营、舞台调度、长跑、登山等，兴趣爱好广泛，**体质测试合格**，具有良好的身体素质；
- **学生任职**：连续四年任**班级班长**，任职中山大学校学生会文体部、中山大学集成电路学院学生会**主要干事**，任中山大学集成电路学院芯火科创协会第一届**副会长**，任中山大学文化建设协会**新媒体部负责人**，任中大青年传媒**新媒体部负责人**，具有良好的任务统筹整理与人员管理调度能力；
- **工作成果**：曾作为梅州市大埔县“红色交通线”暑期调研实践活动的**队长**及**优秀个人**，作为学院2023年艺术文化节**总负责人**获**校精品立项**，作为学院科技文化节**总负责人**策划**2024年科技文化节**，作为主要负责人之一负责**学院三周年文化形象视频**制作并于各大场合播放，有丰富的兴趣技能及良好的活动组织与紧急应变能力。

感谢您的阅读，期待与您交流！本人承诺，简历内容真实准确。如有虚假，愿承担相应责任。

简体中文 | English

Tongxin Zhou

13396210825 | zhoutx8@mail2.sysu.edu.cn

Personal Information

Gender: Male **Date of Birth:** February 2003 **Place of Origin:** Heze, Shandong Province, P.R.China
Undergraduate Institution: School of Integrated Circuits, Sun Yat-Sen University
Political Affiliation: Communist Party of China(CPC) Member
Major: Microelectronics Science and Engineering (Class of 2021) **GPA:** 86.17 **Class Rank:** 9/87(15%)

- **Language Proficiency:** Native in **Mandarin Chinese**; Proficient in **English** with excellent listening, speaking, reading, and writing skills. Achieved **CET-4 (550)** and **CET-6 (485)** scores, with good performance in **CET-6 oral communication**.
- **Core Courses:** Programming, Linear Algebra, Advanced Mathematics, Theoretical Mechanics, Complex Analysis, University Physics, Analog Circuits, Digital Circuits, Mathematical Physics Methods, Signals and Systems, Analog Integrated Circuit Design, Digital Integrated Circuit Design, Quantum Mechanics and Solid-State Physics, Hardware Description Language and FPGA Design, Electrodynamics, Semiconductor Physics, Microcomputer Principles and Embedded Systems, Artificial Intelligence and Hardware Acceleration, Advanced Micro- and Nanoelectronic Devices, Digital Signal Processing, and Electronic Design Automation.
- **Research Interests:** AI and hardware acceleration solutions (**FPGA**), integrated circuit chip architecture design (**neural networks**), MEMS micro-system design, and in-memory computing.

Honors & Awards

- November 2024: **Third-Class Outstanding Student Scholarship, Sun Yat-sen University**
- September 2024: **Outstanding Core Member, ChipFire Science and Technology Innovation Association**
- August 2024: **Second Prize, South China Region, 8th National College Student IC Innovation and Entrepreneurship Competition**
- December 2023: **National** Scholarship
- November 2023: **First-Class Outstanding Student Scholarship, Sun Yat-sen University**
- November 2023: **First Prize, College Career Planning Competition, Sun Yat-sen University**
- November 2023: **Outstanding Individual, "Three Rurals" Summer Social Practice Campaign, Sun Yat-sen University**
- August 2023: **Second Prize, Guangdong Province, National College Student Electronics Design Contest**
- November 2022: **Special Scholarship, Sun Yat-sen University**
- October 2022: **Second Prize, Guangdong Division, National College Student Mathematical Modeling Contest**

Research Experience

Construction and Optimization of a Chinese Large Language Model Based on Parameter Fusion (Undergraduate Thesis)

- **Project Overview:** LLM Merging: Building LLMs Efficiently through Merging shows that **LLM merging** has developed into an efficient way of building LLMs. We are looking for the merging results of **Chinese models**, e.g., ChatGLM, QWen + ShenNong, QWen+VLA, etc.
- **Main Responsibilities:** Conducted merging **tests and optimization** experiments **independently** under the guidance of the advisor.
- **Achievements:** Completed multiple rounds of fusion tests and are progressively **optimizing** model performance.

LAV: Design of a LoongArch-Based Accelerator for R-MSFM Models (8th National College Student IC Innovation and Entrepreneurship Competition)

- **Project Overview:** Proposed a highly customized **hardware architecture** for **visual network inference accelerators (LA-Vision)** based on the **LoongArch architecture**.
- **Main Responsibilities:** Involved in **coding**, **circuit design**, **board debugging**, **simulation testing**, and **live demonstrations**.
- **Achievements:** Passed the initial screening in the South China region and won **the Second Prize** after on-site demonstrations.

Design of a Depth Estimation Hardware Processor for Robotic Vision Systems (Undergraduate Innovation and Training Program)

- **Project Overview:** Designed a **hardware processor** on **FPGA** for **machine-learning-based monocular depth estimation** in robotic vision applications.
- **Main Responsibilities:** Tested and selected **quantization strategies** for the model and developed **hardware acceleration solutions**.
- **Achievements:** The project has been approved at **the university level**, and a multi-layer depth estimation hardware processor has been successfully constructed. Currently, efforts are focused on **real-time** data processing, with plans to achieve real-time depth estimation on the edge.

Motion Target Control and Tracking System (National College Student Electronics Design Contest)

- **Project Overview:** Participated in the "Problem E" of the 2023 competition, designing a system to perform functions such as reset, winding, path following, and tracking via button controls.
- **Main Responsibilities:** Programming, circuit design, and report writing.
- **Achievements:** Successfully implemented all required functions and tested the device at the competition venue, earning the Second Prize in the Guangdong region.

Chemical Composition Analysis Model for Ancient Glassware Based on K-Means Clustering (National College Student Mathematical Modeling Contest)

- **Project Overview:** Tackled "Problem C" in the 2022 competition, analyzing chemical composition data of ancient glassware using **modeling** techniques.
- **Main Responsibilities:** Served as **team leader**, studied principles, designed code, and drafted reports.
- **Achievements:** Solved the problem of chemical composition analysis and won **the Second Prize in the Guangdong region**.

Low-Power High-Resolution Delta-Sigma Modulator for MEMS Capacitive Accelerometers (Undergraduate Innovation and Training Program)

- **Project Overview:** Conducted research in 2022 on Delta-Sigma technology to meet the precision requirements of MEMS capacitive accelerometers with **micro-watt-level power consumption**.
- **Main Responsibilities:** As **team leader**, handled parameter modeling, circuit design, report writing, and final defense.
- **Achievements:** Successfully concluded the project, achieving an effective resolution of **16 bits**, power consumption of **1.67 μW**, and a quantization SNR of **101 dB**.

Interests & Skills

- **Personal Skills:** Proficient in C, Python, Markdown, Cadence EDA suite, AutoCAD, Altium Designer, Vivado, Proteus, and other related tools.
- **Hobbies:** Video editing, new media operations, stage management, long-distance running, and mountain climbing, demonstrating a wide range of interests and excellent physical fitness.
- **Student Leadership:** Served as **class monitor** for four consecutive years. **Active member** of the Culture and Sports Department, Sun Yat-sen University Student Union. **Core executive** in the Student Union of the Integrated Circuit College, Sun Yat-sen University. **Vice President** of the inaugural ChipFire Science and Technology Innovation Association at the School of Integrated Circuits, Sun Yat-sen University. **Head of the New Media Department** for the University's Cultural Construction Association. **Head of the New Media Department** for Zhongda Youth Media. Gained strong organizational and personnel management skills.
- **Achievements:** Team Leader and Outstanding Individual in the "Red Revolutionary Transport Line" summer research and practice activity in Dapu County, Meizhou. **Key Executive** in Production of College's 3rd Anniversary Cultural Image Video, showcased in major events. Led **the 2023 College Arts and Culture Festival as the General Coordinator**, earning recognition as a university-funded boutique project. Planned **the 2024 Technology and Culture Festival** as **the General Coordinator**, showcasing excellent event organization and crisis management abilities.

Thank you for your consideration. Looking forward to collaboration.
I hereby declare that the information provided in this resume is true and accurate. I take full responsibility for its authenticity.